SYNOPSIS

CodeQuest: RPG Coding Adventure

INTRODUCTION

Introduction

CodeQuest is a frontend-based RPG-style coding and DSA practice web application designed to make coding practice more interactive and engaging. In this application, the user plays the role of a coding hero who completes different quests and challenges, earns XP, levels up, unlocks new quests and collects badges.

The main purpose of CodeQuest is to combine **learning with gaming** so that students can practice DSA in a more interesting and motivating way. The application runs completely in the browser. CodeQuest SRS

Problem Statement

Learning and practising Data Structures and Algorithms can become repetitive and difficult for beginners to continue regularly. CodeQuest addresses this problem by providing a game-like environment where coding practice is divided into quests and challenges.

Objectives

- Make DSA learning interactive and engaging.
- Provide beginner-friendly coding challenges.
- Motivate users through XP, levels and badges.
- Track coding progress and daily streaks.
- Provide a simple and responsive user interface.

Target Users

- DSA beginners
- College students
- Coding practice learners
- Students preparing for placements

PROPOSED SYSTEM

Proposed System

CodeQuest provides a structured RPG-style learning journey:

Landing Page → Dashboard → Quest Map → Difficulty → Challenge → Result & Progress

The user starts the adventure, selects a quest, chooses a difficulty and attempts a predefined coding challenge. After submission, the answer is validated and the user's XP, level, progress, streak and achievements are updated. CodeQuest SRS

Main Features

- **Player Profile:** Create a hero profile with name and avatar.
- **Dashboard:** Shows level, XP, streak, completed challenges and current quest.
- **Quest Map:** Contains DSA quests such as Arrays, Strings, Searching, Sorting, Linked List, Stack, Queue and Basic Algorithms.
- **Difficulty Levels:** Easy, Medium and Hard challenges.
- **Challenge System:** Users attempt predefined coding/DSA questions.
- **XP & Level System:** Users earn XP after successfully completing challenges.
- **Achievements:** Badges are awarded for completing specific milestones.
- **Progress Tracking:** Displays completed challenges, quest progress and streak.
- **Local Storage:** Progress can optionally be saved in the browser. CodeQuest SRS

Expected Outcome

The project will provide a simple and engaging platform where students can practise DSA through an RPG-style experience instead of traditional question-solving alone.

TECHNOLOGY & SCOPE

Technology Stack

Technology	Purpose
HTML5	Web page structure
CSS3	Styling and responsive design
Vanilla JavaScript	Logic and interactivity

Technology	Purpose
Figma	UI/UX design and prototype
GitHub	Version control and hosting

The current system is **frontend-only** and does not use a backend, database, framework or external API. CodeQuest SRS

System Architecture

```
USER
  ↓
HTML5
  ↓
CSS3
  ↓
VANILLA JAVASCRIPT
  ↓
CHALLENGE VALIDATION
  ↓
XP + LEVEL + PROGRESS
  ↓
LOCAL STORAGE
```

Development Phases

1. Requirement Analysis
2. Figma UI/UX Design
3. HTML Development
4. CSS & Responsive Design
5. JavaScript Functionality
6. Challenge Integration
7. XP & Progress System
8. Testing
9. GitHub Deployment CodeQuest SRS

Scope

In Scope:
Landing page, dashboard, player profile, quest map, DSA challenges, difficulty levels, XP, levels, badges, streaks, progress tracking, responsive UI and local progress storage.

Future Scope:
Backend, database, login/signup, cloud synchronization, leaderboard, multiplayer, AI coding assistant and real code compiler. CodeQuest SRS

Conclusion

CodeQuest aims to make DSA practice more engaging by combining coding challenges with RPG elements such as quests, XP, levels, badges and progress tracking. The project provides a simple frontend solution that can be developed using HTML, CSS, Vanilla JavaScript, Figma and GitHub.